Technical Supplement

Engineered Intelligence: Tracing Computational Design from Search to Evidence

Research draft, 20 October 2026. Author approval pending.

This supplement retains the detailed representation, score, numerical-method, and cohort inventories supporting the main paper. Original evidence and contracts retain their own producer identities. It reports computational records; physical observations are not reported in this study.

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I. Historical Representation and Search Details

This supplement binds detailed decoding and variation statements to historical producer 82f450f7. The historical accession receipt binds qd.rs at that revision. The associated genome.rs and pheno.rs are inspected source, not evidence that every other cohort used their rules or that the original numerical work was replayed. Manuscript source and evidence-producer revisions remain separate identities.

A. Schema and Literal Producer Domains

The numeric groups occupy indices $0..39$, $40..89$, $90..116$, and $117..127$; the categorical material field follows them. The historical population CSV has 17 metadata/metric columns followed by these 129 genome-and-material fields. Its allele names and bounds are preserved below directly from the producer. A listed domain is a stored-coordinate domain, not an assertion that every value yields a distinct active geometry.

The silhouette bounds are symmetric about zero. Stations 0, 39, and 19–22 are pinned; stations 1–7, 14–18, and 23–27 allow ± 3.0 mm; station 8 and stations 28–29 allow ± 2.0 mm; stations 9–13 and 30–35 allow ± 3.5 mm; stations 36–38 allow ± 2.5 mm.

In the literal tables, T_MIN=5.3 mm, P1_POCKET_MAX=0.6 mm, and P1_RUNOUT_MIN=3.0 mm. The stock ceiling t_max() depends on JARAMBIT_T_MAX, constrained between the floor and 6.0 mm, defaulting to 6.0 mm. The source table alone does not establish a particular run's environment value. Units and threshold/floor interpretation follow the named decoder; source identifiers are retained to avoid inventing unit assignments for ratios and selectors.

TABLE I
Historical literal coordinate domains, indices $40..55$.

Index	Source name	Stored bounds
40	blade_drop_deg	[-4.0,4.0]
41	grip_dx	[-0.4,10.0]
42	ring_wall	[4.5,7.0]
43	ring_id	[24.9,25.9]
44	neck_fillet	[8.0,18.0]
45	t_stock	[T_MIN,t_max()]
46	t_tip	[2.6,4.5]
47	taper_x0	[115.0,150.0]
48	edge_angle_in	[14.0,22.0]
49	edge_angle_sp	[14.0,22.0]
50	comet_x0	[22.0,30.0]
51	comet_x1	[56.0,70.0]
52	comet_w0	[7.0,12.0]
53	comet_w1	[3.5,6.0]
54	comet_bow	[0.0,4.0]
55	comet_y	[0.5,4.0]

B. Clamping and Decoded Types

The historical clamp replaces nonfinite values with lower bounds, applies box bounds, and quantizes stock through the bound q01 routine. It enforces $t_{\mathrm{tip}} \le t_{\mathrm{stock}} - 0.5$ mm, holder-wall and engagement constraints, a fuller interval separation of 0.15 (capped at 0.9), a comet span of at least 28 mm, and a window interval separation of 0.2 (capped at 0.92). These are producer-specific rules, replacing the manuscript's earlier unbound generic examples.

The continuous-rail layer activates at p1_on ≥ 0.5, or through its environment override. It disables selected legacy pocket/through-feature controls, caps shallow pockets, and bounds thickness knots. With the layer off, all stored knot values are overwritten by stock thickness during clamping. Active knot count is floor(k_n) bounded to 3–5. The phenotype uses floor(win_n) for window count and a threshold at 0.5 for the texture mode. It does not round every selector to an integer in the serialized vector. Thresholded zone controls and inactive values further separate the number of stored coordinates from effective geometric changes.

TABLE II
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES $56..71$.

Index	Source name	Stored bounds
56	eye_scale	[0.0,1.25]
57	cairn_d	[4.0,7.0]
58	nub_engage	[0.4,1.2]
59	holder_wall	[1.2,2.0]
60	jimp_scale	[0.8,1.2]
61	serr_large_d	[1.0,1.5]
62	serr_small_d	[0.4,0.8]
63	serr_fine_d	[0.5,0.9]
64	spine_sharp_offset	[0.0,45.0]
65	relief_d	[0.0,1.2]
66	relief_w	[2.0,8.0]
67	relief_q	[1.0,3.0]
68	lattice_d	[0.0,1.5]
69	lattice_pitch	[4.0,9.0]
70	lattice_frac	[0.4,0.8]
71	lattice_angle	[0.0,60.0]

TABLE III
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES $72..87$.

Index	Source name	Stored bounds
72	lattice_handle	[0.0,1.0]
73	lattice_blade	[0.0,1.0]
74	fuller_d	[0.0,1.2]
75	fuller_w	[2.0,5.0]
76	fuller_s0	[0.05,0.4]
77	fuller_s1	[0.5,0.9]
78	flute_d	[0.0,1.0]
79	texture_mode	[0.0,1.0]
80	texture_d	[0.15,0.4]
81	texture_pitch	[1.5,3.0]
82	texture_angle	[-45.0,45.0]
83	wrap_x0	[16.0,26.0]
84	wrap_len	[45.0,60.0]
85	pad_x	[70.0,80.0]
86	pad_y	[-5.0,3.0]
87	pad_w	[6.0,9.0]

C. Crossover and All Six Mutation Branches

Historical crossover chooses the 40-coordinate silhouette block from either parent with equal probability, then chooses every other numeric coordinate and the material category independently with equal probability, followed by clamping.

For mutation, op=floor(6u) selects a mode from 0–5 using the producer RNG. The supplied scale is σ; fine means $\sigma < 0.025$, and silhouette scale becomes 0.1σ in that branch. Let s_i denote coordinate-domain span. A jig adds a normal variate multiplied by the supplied scale and s_i.

1) Mode 0: attempt each coordinate with probability 0.15, using silhouette scale for the first 40 and σ otherwise; stock is omitted in fine mode.
2) Mode 1: choose a periodic silhouette-bump center in $[0, 40)$, width from $[1.5, 5.0]$, and normal amplitude scaled by 6 times the silhouette scale. Apply the source's exponential profile with station-bound scaling. Outside fine mode, also attempt drop and grip-offset perturbations, each with probability 0.3.
3) Mode 2: attempt thickness, taper, relief, bevel, ring-wall, and edge-thickness controls with probability 0.5 and jig scale 1.5σ, excluding stock in fine mode. When the rail layer is active, attempt each of five knots and knot count with probability 0.5.

TABLE IV
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES 88..103.

Index	Source name	Stored bounds
88	pad_l	[12.0,16.0]
89	edge_t	[0.5,0.9]
90	iso_d	[0.0,1.6]
91	iso_pitch	[7.0,16.0]
92	iso_rib	[1.4,3.0]
93	iso_angle	[-30.0,30.0]
94	iso_rim	[2.0,5.0]
95	iso_blade	[0.0,1.0]
96	iso_handle	[0.0,1.0]
97	iso_blade_extent	[0.0,1.0]
98	iso_grade	[-0.8,0.8]
99	win_n	[0.0,7.99]
100	win_s0	[0.05,0.5]
101	win_s1	[0.5,0.92]
102	win_w	[3.0,10.0]
103	win_fill	[0.45,0.8]

TABLE V
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES 104..119.

Index	Source name	Stored bounds
104	win_decay	[0.4,1.0]
105	rflute_d	[0.0,2.0]
106	rflute_w	[0.3,0.7]
107	rflute_arc	[120.0,260.0]
108	iso_handle_start	[0.0,0.8]
109	iso_land	[0.0,1.0]
110	perf_d	[0.0,7.0]
111	perf_gap	[3.2,8.0]
112	perf_angle	[-40.0,40.0]
113	perf_elong	[1.0,2.4]
114	perf_land	[0.0,1.0]
115	perf_handle	[0.0,1.0]
116	perf_handle_start	[0.0,0.9]
117	p1_on	[0.0,1.0]
118	rail_w	[3.0,7.0]
119	neck_l	[4.0,12.0]

TABLE VI
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES 120..127.

Index	Source name	Stored bounds
120	pocket_d	[0.0,P1_POCKET_MAX]
121	pocket_r	[P1_RUNOUT_MIN,6.0]
122	k_n	[3.0,5.99]
123	k0	[T_MIN,6.0]
124	k1	[T_MIN,6.0]
125	k2	[T_MIN,6.0]
126	k3	[T_MIN,6.0]
127	k4	[T_MIN,6.0]

4) Mode 3: attempt lattice depth, pitch, fraction, and angle with probability 0.6 at scale 1.5σ. Outside fine mode, flip each handle/blade activation coordinate with probability 0.2.
5) Mode 4 (the remaining match branch): attempt fuller, flute, texture, comet, eye, and cairn controls with probability 0.4 at scale 1.5σ; omit texture mode in fine mode.
6) Mode 5: with the rail layer active, attempt rail, neck, pocket, knot, stock, fuller-depth, ring-wall, and neck-fillet controls with probability 0.4 at scale 1.5σ. Otherwise attempt the source's machining-coordinate list with probability 0.4; outside fine mode, flip five zone controls with probability 0.12 and attempt a ± 1 window-count step with probability 0.25. Stock is omitted in fine mode.

Outside fine mode the material category flips with probability 0.03. Every mutation ends in clamping. The historical island loop supplies island/configuration-dependent scales, including a 0.6 multiplier after crossover; the earlier manuscript's fixed 0.15/0.04 pair did not describe this bound loop. Source lists remain available in the immutable genome implementation.

D. Membership Stage and Frame Interpretation

The bound island loop constructs novelty context, refreshes fitness, emits birth/elite membership events and summary observations, and calls its snapshot callback before migration. Migration copies selected IDs to the next island. A GenEnd event is emitted after that migration, but its summary values were gathered before it. Event position and the stage summarized by its values are therefore different facts. JEV1 elite events omit refreshed scalar fitness. The reporting adapter accepts complete and maps historical notevaluated to not_evaluated; it rejects unknown status strings.

E. Inheritance Ledger and Interpretation Boundary

Two source-bound reporting paths coexist. The original lineage-inputs.v1.json defines a two-hop view consumed by lineage::project, which produces generated/lineage.json and generated/lineage.svg. Separately, the population reporter reconstructs generated/population/inheritance.json directly from the hash-pinned historical all_evals.json and all_evals.csv. Its validator checks the seven three-hop parent edges and separately identifies the recorded fourth-hop induced closure $174 \to 299$; this path produces figures/inheritance.svg, the ancestry view used here. The ledger also retains exact numeric-coordinate differences, material change indicators, and mixed-unit raw norms. The two-hop contract remains unchanged and is not the authority for the separate three-hop view.

II. Scores, Candidate Roles and Display Composites

A. Original Reporting Authority and Missing Provenance

The six sources admitted by docs/paper/weighted-inputs.v1.json contain fifteen generation summaries. That original contract leaves producer revision, scorer revision and point stage unestablished. The original score-profile registry also names a producer tuple for the weighted profile: revision 009f5ac520fa3afe8ed8025230eef604dbb7f922, path crates/paper-artifacts/src/validation.rs. That path is absent at the named revision. Preserve the original registry and this unresolved binding; the saved vector below does not reconstruct the missing historical scorer. Matching metric names or current source cannot establish an executed historical revision.

The source for Individual 2002's vector is evidence/evolution_waves/2026-10-18_ultralight_v1/xbabe3/w411c3/fitness_curve.json, SHA-256 afe065170218f0f5f20942a9e737e7e91a300244f8e6a99fd3c1d617d16f1412, selector /best_so_far/ultralight_fitness_breakdown/objectives. The scalar is 74.207299. The weights and normalized terms below are serialized summary values; they are not recomputed solver observations. Their weighted-point sum exactly reproduces the saved scalar to its retained precision. No CAD geometry or independent later assessment is inferred from the registry category, certificate hash or component labels.

TABLE VII
INDIVIDUAL 2002'S ACTUAL RETAINED WEIGHTED VECTOR: SCIENTIFIC LABELS PLUS EXACT MACHINE KEYS. NORMALIZED VALUES AND CONTRIBUTIONS ARE SAVED REPORTING QUANTITIES.

Objective / machine key	**Weight (%)**	**Normalized**	**Points**
Mass preference; ultralight_mass	22.0	0.339548	7.470061
Holder/compactness proxy; holder_and_compactness	8.0	1.000000	8.000000
Ring/claw form proxy; karambit_ring_and_claw	7.0	1.000000	7.000000
Reference form proxy; c6_aesthetic_and_ergo	9.0	1.000000	9.000000
Bore/collar comfort proxy; bore_collar_comfort	9.0	0.875000	7.875000
Single-solid B-rep status; cad_watertight_brep	5.0	1.000000	5.000000
Exported edge-profile proxy; gate_s_sharpness	5.0	0.846429	4.232143
Exported point-profile proxy; gate_s_point_profile	5.0	0.521709	2.608543
Certification lateral, 180 N; hiker_lateral_180n	3.5	1.000000	3.500000
Certification lateral, 260 N; hunter_lateral_260n	2.5	0.708077	1.770192
Strict lateral, 260 N; strict_hunter_260n	2.0	0.757692	1.515385
Ring traction, 1500 N; ring_pull_1500n	2.5	1.000000	2.500000
Body torsion, 1.82 N m; twist_body_1p82nm	2.5	1.000000	2.500000
Local plate torsion, 1.82 N m; twist_tip_plate_1p82nm	2.5	0.260000	0.650000
Reported solid-tip/comparison proxy; solid_tip_3d_p2_and_xcheck	3.0	0.363600	1.090800
Reported support/stability proxy; side_a4_buckling	2.0	1.000000	2.000000
Elastic yield-order proxy; failure_sequence_1p2	3.5	1.000000	3.500000
Reported plate-study status; plate_gci_3level	2.0	1.000000	2.000000
Overload proxy; human_max_overload	2.0	0.483333	0.966667
Goodman fatigue proxy; goodman_fatigue_l7	2.0	0.514254	1.028508
Total	100.0	—	74.207299

B. Conditional CAD Score and Global Composite

At producer revision e8cb049df7d03c7d3113556812cc140ca99a29c3, ops/ci/evaluate_cad_candidate.py defines the CAD score's two branches:

$$W_{\mathrm{CAD}}^{\mathrm{pass}} = 1 + 0.70 s_{\mathrm{mass}} + 0.12 s_{\mathrm{form}} + 0.10 s_{\mathrm{survival}} + 0.05 s_{\mathrm{seq}} + 0.03 s_{\mathrm{carry}}, \tag{1}$$

$$W_{\mathrm{CAD}}^{\mathrm{fail}} = -\left(1 + 4 \sum_{k} \Delta_k\right) + 0.10 s_{\mathrm{mass}}. \tag{2}$$

The pass branch applies only if all sixteen legacy predicates are true. The mass subscore is a clipped mixture of linear and logarithmic rewards between the 92 and 74 g references. Form combines reference score, hook-drop and curvature proxies. Survival and sequence terms use saturating margin functions; the carry term combines the saved support eigenvalue and fatigue factor. These historical names and weights describe the score's preference rule; they do not establish the physical interpretation implied by a component name.

For Individual 2000, the bound scorecard saves $s_{\mathrm{mass}} = 0.486999$, $s_{\mathrm{form}} = 0.400862$, $s_{\mathrm{survival}} = 0$, $s_{\mathrm{seq}} = 0.999604$ and $s_{\mathrm{carry}} = 0.6$. Its nonzero normalized shortfalls are strict lateral 0.3530769231, strict tip torsion 0.471 and solid tip 0.3307. The saved penalty is 4.619108 and the actual failed-branch score is -5.570408. The saved unpenalized value 1.456983 is hypothetical for this failed candidate. Favorable legacy flags such as a plate-study metadata predicate remain stored bookkeeping; they do not override the refused extrapolated-error estimate or open eigen audit.

The separate dimensionless composite is

$$G_1 = 0.35 \min(\mathrm{SF}_{\text{solid tip}}, 2) + 0.35\,\mathrm{SPG}/4 + 0.20 s_{\mathrm{shape}} - 0.05 \max(0, m_g - 77)/10 - 0.05 \max(0, 1.2 - r_{\mathrm{seq}})/1.2. \tag{3}$$

The original metrics/global_metric.py requires valid inputs rather than filling missing terms with zero. A screen-tip substitute is explicitly provisional. A complete G_1 status concerns input completeness; it does not mean the candidate satisfies all requirements. Individuals 2001/2007 retain $G_1 = 0.623903/0.621634$ and $W_{\mathrm{CAD}} = -3.878704/-3.964924$ under their separate controlled-export records.

C. Native Context and Display Transform

The historical native scorer is bound to 82f450f7fb418e28c4b229031920dec5454435f3, crates/crescent/src/qd.rs, SHA-256 fe3d057eeec74c3278b46027265406ec0ee97dd65938ae49cb003e08ee5b3a5f. Illegal/infeasible branches can be negative; evaluated records add mass/mechanical terms and contextual novelty. Novelty uses contemporaneous descriptor context and is re-evaluated during search. A legal geometry can still have negative fitness. Stored birth observations, rescored memberships, generation/island summaries and cumulative archive representatives are distinct stages, even when they refer to the same individual.

Negation $L = -f$ or $-W_{\mathrm{CAD}}$ and complement $L = 100 - F$ reverse numerical order while preserving preference under minimization. A cumulative maximum $H_g = \max_{k \le g} r_k$ is nondecreasing by construction; its loss transform is nonincreasing. These transforms preserve a profile's preference, not comparability between profiles or evidence of physical improvement.

D. Actual Field-Ranking Display Recipe

The retained hero implementation sorts active cells within each field and assigns cell i the area-interval midpoint $(A_{<i} + a_i/2)/A_{\mathrm{active}}$. Equal serialized values are visited in deterministic source-cell order, not pooled into a single tied-value rank. It averages those ranks over active maps per cell, then applies a candidate-specific affine stretch from the resulting minimum/maximum to $[0, 1]$. Thus its final color index is a stretched relative rank summary, not a direct stress, a calibrated quality probability or a common absolute percentile across candidates.

The five maps are envelope, lateral, pry, ring and raw torsion. The envelope and ring use producer-active domains; lateral/pry/torsion retain all cells. The source support is 10,334 cells with five contributors, 14,800 with four and 75 with three. The cover does not display a separate contributor map. The envelope depends on the primitive responses; the saved primitive-only comparison reports area-weighted L_1 index difference 0.030693 and maximum difference 0.122172 relative to the five-map recipe. These saved sensitivity values describe a display comparison and are not independent mechanical evidence.

E. Crop Measure, Denominator and Range

For a piecewise-constant stored triangle field v, the display-only saliency measure is $\int_D |v|\, dA$. A crop's numerator integrates over the actual triangle/bbox intersections, and its denominator integrates over all source-active triangles in the declared domain. A chosen prior ROI is a different denominator from the whole retained field. Coordinates use the proper rigid pose $R_z(-6^\circ)R_x(180^\circ)$.

The following is a literal-declared-bbox diagnostic, not an achieved-coverage claim for the rendered SVG viewport. Aspect-ratio padding can enlarge the actual clip domain beyond the printed bbox, and emitted polygon counts must be source-bound separately. For the shipped printed bounds, the ring crop is $[-28, -26, 84, 23]\,\mathrm{mm}$, the terminal torsion crop $[166, 4, 198.5, 37]\,\mathrm{mm}$ and the pry view $[-20, -26, 196, 36]\,\mathrm{mm}$. Polygon-clipped reconstruction from the original serialized cells gives active whole-domain integrals 14.173756, 119.158584 and 900.438870, respectively. The literal-bbox diagnostic integrals are 14.173756, 76.878008 and 900.438870, giving 100%, 64.5174% and 100%. These fractions do not establish actual visible-domain coverage. Final emitted-output receipts must bind each drawn polygon, affine viewport, clip domain and denominator to the actual SVG/PNG before any achieved-fraction caption is accepted.

The superseded terminal receipt instead reports 2062 selected ROI cells (not an established count of currently emitted polygons) and integral 73.5323, compared with whole torsion integral 119.1586 over 25,209 cells. Its 100% ROI fraction is only about 61.71% of that whole-field denominator. Its exact prior ROI selection rule is unstated; it is not re-created or certified here. Preserve that receipt with explicit supersession rather than updating its hashes around a changed panel.

Raw torsion peaks at $34.8589\,\mathrm{MPa}/(\mathrm{N\,mm})$, inside the capacity exclusion; the post-exclusion capacity peak is $0.6611\,\mathrm{MPa}/(\mathrm{N\,mm})$. A view using the latter legend range needs explicit over-range labeling and a full-range

TABLE VIII
RETAINED SOLID IMPLEMENTATION CHAIN. UNKNOWN EXECUTED PACKAGE VERSIONS REMAIN UNKNOWN. RUST IS THE PIPELINE/ORCHESTRATION LAYER, NOT EITHER SOLID ASSEMBLY LIBRARY.

Property	Primary	Comparison
Producer	fem_solid/tip_solid.py	fem_solid/cross_check.py and shared DOLFINx seams
Assembly/field	scikit-fem ElementVector(ElementTetP2), integration order 2	DOLFINx vector Lagrange P_2 on linear tetrahedral geometry
Linear solution	PyAMG smoothed aggregation, CG; recorded algorithm skfem_p2_amg_cg	PETSc CG/GAMG; Chebyshev/Jacobi multigrid-level options
Retained versions	scikit-fem and PyAMG package versions not retained	DOLFINx 0.9.0; PETSc/petsc4py 3.22.0; Basix 0.9.0; UFL 2024.2.0
Convergence record	152 iterations; relative residual 8.803574×10^{-11}	158 iterations; positive KSP reason 2; sidecar residual norm 1.324748×10^{-10} (a different reported norm)
Common mesh	Saved NPZ with $91{,}008$ tetrahedra and $416{,}583$ displacement degrees of freedom; mesh-file SHA-256 d3f7a444175fb6a6f64ee1118a00f2e83d6994c66b1c266cfd94ac14572faab1. This file identity differs from the sidecar's internal mesh-layout hash.	
Material/load	Ti-6Al-4V isotropic elasticity, $E = 113.8\,\mathrm{GPa}$, $\nu = 0.34$; common saved allowable and $1820\,\mathrm{N\,mm}$ chord torque. Saved primary and comparison scalar outputs use different rounding.	
Boundary	Zero displacement on cut-arc vertices and midpoint degrees of freedom on edges with both ends fixed. Minimum-norm z-forces at vertices within $2\,\mathrm{mm}$ of the tip impose torque with zero resultant force and zero transverse moment. No transferred-displacement boundary is recorded.	
Recovery/region	Element-vertex von Mises stress, then incident-cell vertex maximum or unweighted mean; planar tip distance in $[4, 15)\,\mathrm{mm}$ and nearest saved cut-vertex distance $> 3\,\mathrm{mm}$. The comparison sidecar records 6930 ROI vertices.	

reference. Neither silent saturation nor a caption alone can supply that missing visual key. Panel C is always /mesh/pry, peak $2.9181\,\mathrm{MPa/N}$, rather than the envelope in MPa.

III. NUMERICAL METHODS, SOURCE BINDINGS AND OBSERVATION LIMITS

This supplement interprets retained observations for Individual 2000; it reports no new solve. The original field, assessment and verification input contracts bind the producer revision e8cb049df7d03c7d3113556812cc140ca99a29c3. The producer source and raw-input hashes must be checked against those original contracts. A later figure-semantic ledger describes presentation and cannot independently certify a scientific interpretation.

A. Actual Solid Implementation Chain

The primary code contains a direct-solver fallback, but the retained run explicitly reports the PyAMG/CG branch. The comparison interpolates von Mises stress into DG1 at cell vertices; it is not an L_2 projection, volume weighting or nodal smoothing. Shared incident-cell recovery produces the reported scalar extrema. Differences computed from rounded safety factors are $|0.6693/0.669 - 1| = 0.0004484$ and $|0.7263/0.726 - 1| = 0.0004132$. The code binding and these values support the selected comparison, not a global proof of solver correctness.

B. Plate Stages, Fields and Regions

The producer's CERT_AREAS are 0.25 and $0.125\,\mathrm{mm}^2$. Certification takes the specified minima over both unbiased solves; its retained spatial fields come from $0.25\,\mathrm{mm}^2$. In the assessment record, lat_rob_n and twist_tipzone_sf therefore denote certification values. strict_lat_rob_0p0625_n and twist_tipzone_sf_strict denote the separate unbiased $0.0625\,\mathrm{mm}^2$ stage. The respective saved pairs are $168.9008053/168.1786057\,\mathrm{N}$ and $0.523920871/0.528698267$.

Lateral and torsion fields use DKT bending stresses recovered as a maximum over three extrapolated corners. Pry and ring fields use T6 centroid von Mises stress. In particular, /mesh/pry has $25{,}209$ stored unit-load values and peak $2.9181\,\mathrm{MPa/N}$. Its producer applies a unit in-plane load perpendicular to the ring-to-tip chord with the ring

bore clamped. It must not be labeled as the multi-case envelope in MPa. The original field contract's display-field list omits this array; admission requires an explicit additive producer-backed interpretation while preserving the original contract bytes.

The envelope scales lateral/pry fields by $260\,\mathrm{N}$, ring by $1500\,\mathrm{N}$ and raw torsion by $1820\,\mathrm{N\,mm}$ times the producer's global torsion correction. It masks 75 tip cells and has peak $1388.6695\,\mathrm{MPa}$. Raw torsion retains its $34.8589\,\mathrm{MPa/(N\,mm)}$ tip-patch peak; the post-$4\,\mathrm{mm}$ exclusion capacity peak is $0.6611\,\mathrm{MPa/(N\,mm)}$. Body torsion searches beyond $15\,\mathrm{mm}$, while the tip-annulus observable searches 4–$15\,\mathrm{mm}$. No strict spatial field or full volumetric solid stress array is retained in the publication inputs.

C. Three-Level Study and Refusal Semantics

The independent plate study has maximum areas 4, 1, $0.25\,\mathrm{mm}^2$, with $h = \sqrt{A_{\max}}$ and refinement ratio 2. In fine-to-coarse order, the yield-order proxy is 1.4381383, 1.5763238, 1.5948326 and the lateral safety factor is 0.6535320, 0.6886658, 0.6810133. The stored relative-GCI procedure records valid sequence/observable metadata but refuses the former for nonpositive observed order ($p = -2.9003199$) and the latter for oscillation. Extrapolated values and fine-grid estimates are null. A valid sequence descriptor is not an admitted extrapolation, and refusal is not a universal statement about other error-estimation methods.

D. Other Observation Families

Representation/export records bind genome, material, B-rep, surface mesh, mass and their digests. For Individual 2000, the saved export reports one solid and one shell, a valid OpenCascade B-rep and a $44{,}994$-vertex/$90{,}000$-triangle display mesh. The reference comparison uses a 4230-point exterior contour and 1117-point authored reference; saved distance $2.285\,\mathrm{mm}$, curvature EMD 0.0071 and similarity 0.4079 describe conformity to that prior. Export metrology and nominal coordinates are distinct observations; neither establishes comfort or human aesthetic approval.

The regional sequence proxy takes the governing ratio of regional elastic capacities over lateral, torsion and pry cases, with a stored target of 1.2. It does not model fracture, crack initiation or plastic redistribution. The modified-Goodman scalar obeys $n_f^{-1} = \sigma_a/S_e + \sigma_m/S_u$ with $\sigma_a = \sigma_m = \sigma_{\max}/2$ for the modeled zero-to-maximum cycle; its saved value is 0.47179485. No spatial fatigue array or measured life is supplied.

The alternate support study records radial, tangential and flange springs of $50{,}000$, 5000 and $100{,}000\,\mathrm{N/mm}$ and a Winkler grip foundation of $200\,\mathrm{N/mm}^3$. Its static lateral ratio is 5.2026756; the stored eigenvalue is 7.0350619. Those scalars are conditioned on the alternate supports, rather than replacing the certification ratio 1.4353236. The $\mathbf{K}_s^{-1}\mathbf{G}$ symmetry/eigenpair question remains open. No retained eigenvector is replaced by a synthetic mode shape.

Finally, the historical battery has 23 subjects and 18 columns (414 status cells), including shared software controls, side-study references, blocked geometry bindings and 20 solid-file completeness labels. The original 44-entry crosswalk and historical matrix remain preserved by the publication package. Definition counts (178 assurance challenges/356 paired slots) and repeated control labels are separate from actual execution receipts and from candidate-specific mechanical evidence. A favorable completeness flag is not a favorable strength factor.

IV. Cohort Inventory, Overlap and Descriptive Summaries

The inventory separates original run identity, retained capture and record kind. Receipt observation times below identify accession observations, not independently verified historical OS process start/stop times. Exact raw fields and source precision remain in the immutable JSON accessions.

A. Coverage Sources and Status Qualifications

H: docs/paper/JR_EI_P1_T04_HISTORICAL_RECONCILIATION_RECEIPT_20261019.json reconciles 1984 births, 2176 memberships, 19 event frames and selected later strict records for historical Run 1. Its 146-column CSV retains numeric coordinates/material; its event stream supplies recorded parentage. Selected geometry/fields do not imply per-birth CAD coverage.

P: docs/paper/JR_EI_P1_T05_NEWER_BIRTHS_SCREENS_JOIN_RECEIPT_20261019.json admits 57 proposal/screen file pairs across 24 directories: 3649 proposals, 2259 one-based joins and 1390 missing screens. The

TABLE IX
EARLIER SUMMARY CAPTURES AND OVERLAP WITH COMPLETED RUNS. EXACT PACKAGE IDS ARE RETAINED FOR CROSS-CAPTURE IDENTITY; CAPTURE HASHES ARE IN THE ADMITTED P1-T06/T08 RECEIPTS.

Original package ID	Earlier rows	Generation range	Later completed run
evo_full_20261018_xbabe2_c01	1900	0–474	Not in completed set
evo_full_20261018_xbabe2_c02	64	0–15	Not in completed set
evo_full_20261018_xbabe2_c03	72	0–17	Not in completed set
evo_full_20261018_xbabe2_c04	16	0–3	Not in completed set
evo_full_20261018_xbabe2_c05	2132	0–532	Not in completed set
evo_full_20261018_xbabe2_c06	908	0–226	Not in completed set
evo_full_20261018_xbabe3_c01	2836	0–708	Run 6
evo_full_20261018_xbabe3_c02	2792	0–697	Run 7
evo_full_20261018_xbabe3_c03	2748	0–686	Run 8
evo_full_20261018_xbabe3_c04	2792	0–697	Run 9
evo_full_20261018_xbabe3_c05	1412	0–352	Not in completed set

saved screens lack the later ultralight weighted fitness and remain a separate profile. Recorded proposal metadata and selected exports supply partial parentage/geometry coverage; no all-proposal downstream assessment is inferred.

N: docs/paper/JR_EI_P1_T08_LOSSLESS_GEN800_POPULATION_ACCESSION_RECEIPT_20261019.json admits ten complete 146-column population tables (128 numeric coordinates plus material within 146 total fields), 1,954,560 rows and 200 compressed segments. The original CSVs total 3,811,896,749 bytes; compressed segments total 494,843,141 bytes. The admission receipt records lossless identity. Parent columns are absent from these tables; other retained event streams do not by themselves establish a complete analyzed genealogy. Twenty-four separately retained STL artifacts are inventoried apart from the population payload and do not establish per-birth geometry binding. Later CAD/solid assessments and spatial arrays are not retained per birth in this accession.

E: docs/paper/data/selection_receipts.v1.json binds the editorially illustrated Individual 2000 to the automatic manifest and its original field/assessment/verification contracts. CAD and plate fields are available; full solid, strict plate, eigenvector and fatigue spatial arrays are not publication inputs. This record is separate from the historical 23-subject battery. C: the controlled-probe CAD scorecards bind Individuals 2001 and 2007; their ancestry to a particular evolutionary birth remains unestablished. In all rows, physical observations are not reported in this study. No absence is strengthened to "not performed" without evidence of that fact.

B. Current Native Directories and Earlier Summary Captures

P1-T06 observes its native-summary accession at 2026-10-19T17:41:30Z. P1-T08 records its population accession at 2026-10-19T18:22:30Z. Those are receipt times. The current directory universe has 18 members: ten completed generation-800 directories and eight others. The earlier eleven-file snapshot contains 17,672 summary rows, reaching generation 708. Four original identities occur in both captures. Public Runs 6–9 name those completed runs; they are not four additional replicates merely because earlier summary captures exist.

The eight other current directories are xbabe1/c02, xbabe2/c01--c06 and xbabe3/c05; their retained summaries have differing endpoints. The nine aborted crescent attempts are separately admitted by P1-T03: seven retain 216 live-history rows in total and two retain no CSV rows. Aborted island attempts and their proposal files are another record universe; they are not added to these nine crescent attempts or the completed-population denominator.

C. Ten-Run Descriptive Summary

The run is the unit for these descriptive comparisons. Sample standard deviations use $s^2 = \sum_{r=1}^{10}(x_r - \bar{x})^2/9$; no confidence interval or independence of births is asserted. Native fitness includes contextual novelty, so maxima summarize retained preference under each run's context rather than a common physical-performance measurement.

The six weighted-reporting files contain fifteen summary rows and do not retain their underlying round populations/CAD. Individual 2002's registry category RetainedExport and hashes are preserved as original metadata; that category is not evidence that its export geometry or later assessment is present. Its reporting vector remains

TABLE X
COMPLETED NATIVE RUNS. EACH HAS 195,456 BIRTHS AND 3204 GENERATION/ISLAND SUMMARY ROWS. THE ORIGINAL RECORDS RETAIN ADDITIONAL DISTRIBUTIONS AND EXACT PRECISION.

Run	Search-legal births	Maximum stored fitness	Generation-800 legal median
Run 2	74874	1.1327	-0.2462
Run 3	77011	1.1089	-0.2211
Run 4	77396	1.1380	-0.1933
Run 5	77625	1.1440	-0.2225
Run 6	70278	1.1156	-0.2534
Run 7	74777	1.1228	-0.2023
Run 8	71550	1.1108	-0.2351
Run 9	76284	1.1408	-0.2072
Run 10	74309	1.1186	-0.2302
Run 11	80132	1.1177	-0.2610

separate from Individual 2000's response fields. Original registry/contract contradictions require a scoped additive correction rather than silent historical rewriting.

V. Related-Work Interpretation Notes

The main comparison concerns representation, search, evidence retention, and validation scope. Topology-optimization context [13] establishes that computational changes to structural form are longstanding; no implementation equivalence to every method in that review is asserted. DeepCAD and Text2CAD concern CAD sequences and learned generation [2], [3]. The ST5 preprint supplies a concrete earlier evolutionary-engineering example with fabricated prototypes; its report-specific qualification status is preserved [4]. MAP-Elites supplies archive context [5]; actual descriptors and retained coverage must still be reported for Jarambit's cohorts. FunSearch is limited here to proposal/evaluation separation [6].

PROV-DM's entity/activity/derivation/collection distinctions provide established terms for discussing provenance [14]. The present study does not implement or test a PROV interchange profile. FAIR's discovery and reuse goals and software-citation specificity help separate an accessible research object from a checksum alone [15], [16]. A source snapshot is cited as a snapshot; its identity is not substituted for an unrecorded execution environment or for this manuscript's eventual release identity.

Software methods references support library roles, not a numerical model's validation [8], [9]. The cited DOLFINx v1 preprint from 2023 remains a valid publication object; it differs from the retained DOLFINx 0.9.0 environment. The consulted SciPy API states the symmetric/Hermitian operator contract [11]; its current documentation version does not identify the unavailable historical SciPy version. Roache's grid-refinement reference is retained [12]; the bound reporting procedure and its actual refusal rules, rather than the citation alone, establish which estimates were admitted.

Crameri *et al.* motivate attention to perceptual color gradients and accessibility [17]. This citation does not validate Jarambit's percentile composite, lighting, saliency choices, or final figure accessibility. Those require their own definitions and final visual inspection. The NIST artifact supports distinguishing digital geometry from measured process behavior [10]. NASA's standard supplies credibility context; compliance is not asserted [7].

VI. Complete Historical 23-Subject, 18-Evaluator CAD Battery

A. A Matrix of Different Evidence Kinds

The historical battery contains 23 retained subjects and 18 evaluator columns. Its 414 cells include software capability, geometry binding, numerical findings, shared reports, and record completeness. The manifest fixes membership. A later automatic assessment is a separate corpus and does not silently create a twenty-fourth subject. All cells in this historical corpus report fresh_this_lane:false.

The stored statuses reconcile to 221 passes (221), 81 failures (81), and 112 blocked records (112). These are labels from heterogeneous producers. They are preserved here as historical data alongside the exact explanation of

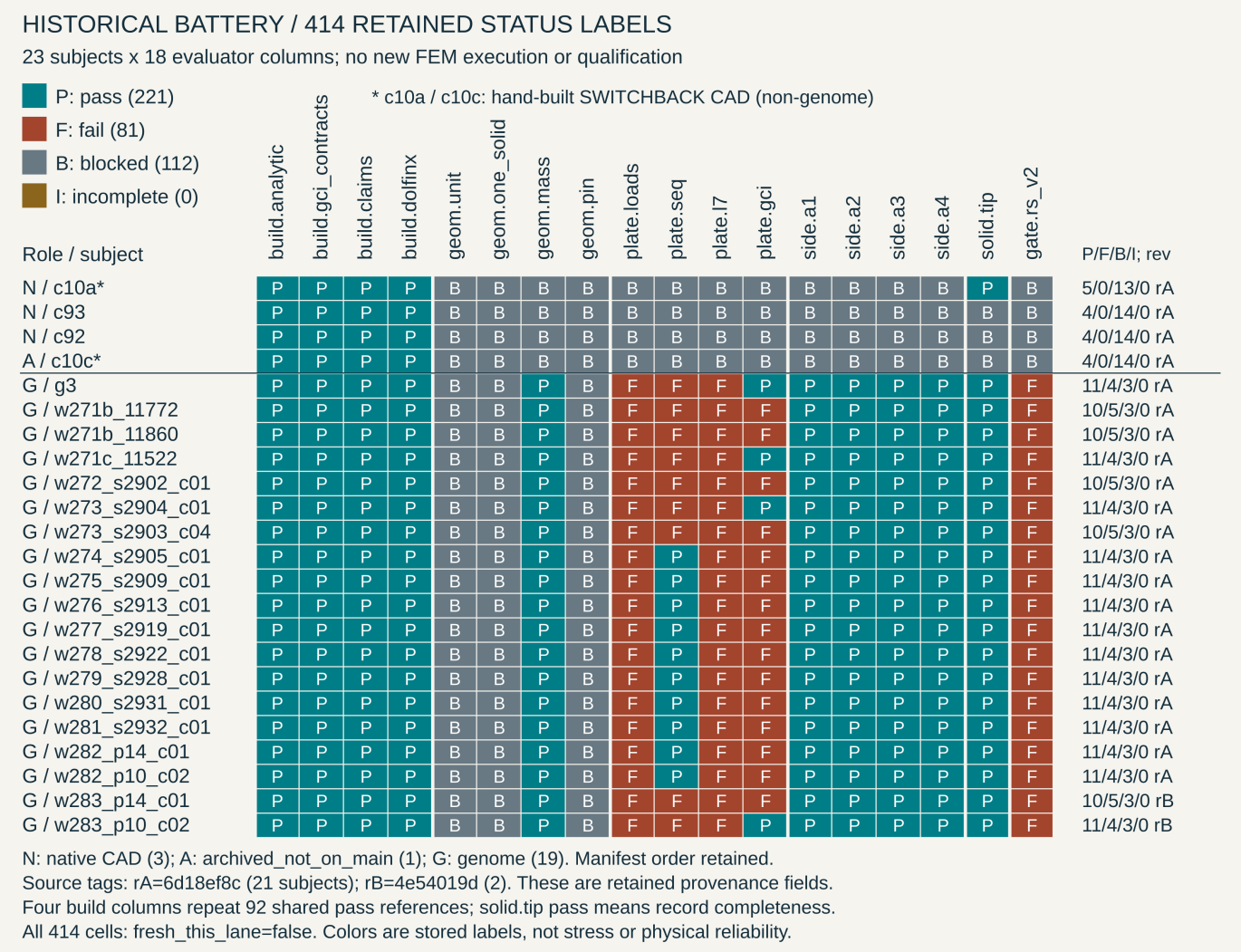


Fig. 1. Supplement Figure S1: All 414 original labels in the fixed historical battery, separate from the cover's assessed example (cert_pipeline_r01_p14_c02). N, A, and G reproduce source roles: native CAD, archived-not-on-main, and genome. P/F/B/I mean stored pass/fail/blocked/incomplete, not common physical outcomes. Source tags rA/rB identify the different retained revision fields (6d18ef8c vs. 4e54019d). Four build columns repeat 92 shared pass references; solid.tip passes mean record completeness. No cell is fresh in this lane, no missing observation is imputed, and no row is promoted to qualification.

what each label meant. In particular, the four shared build columns account for 92 stored passes (4×23), and the local-solid column's 20 favorable statuses concern record completeness. Neither fact supports an aggregate claim of 221 independently successful physical engineering tests.

Figure 1 retains every subject/evaluator pairing in manifest order. Each subsection below refers to the column with its exact evaluator ID and docs/tests/README.md anchor. Letters accompany the colors so missing evidence remains distinguishable from an unfavorable observation. Source-cell selectors, original kinds, and evidence-path strings remain in the linked projection (docs/paper/generated/battery-matrix.json); the aggregate table remains in the artifact package. The manifest's source tag is not uniform across its members: 21 subject records cite 6d18ef8c, while the two w283 records cite 4e54019d. These are retained provenance fields, not a newly verified homogeneous solver regime.

B. Exhaustive Column-by-Column Historical Battery Reference (18 Evaluators)

1) build.analytic: Analytic Software Controls (battery-build.analytic): This column references shared analytic/reference software controls (invariance.rs). Such controls can expose errors in units, invariance, assembly, or solver behavior on reference problems. A per-subject copy of a shared control status does not establish another execution.

The historical 23 passes should therefore be understood as repeated references to capability evidence. They do not establish candidate-specific geometry validity or physical behavior.

2) build.gci_contracts: Convergence Arithmetic (battery-build.gci_contracts): This column concerns software rules for interpreting mesh-study data (test_gci.py). Positive and negative controls demonstrate that a formula or refusal condition is implemented as intended. It does not mean that a particular CAD has a suitable three-level study, matching observables, or an admitted asymptotic error estimate. Candidate-specific convergence belongs to plate.gci.

3) build.claims: Reporting and Admission (battery-build.claims): This column references controls over how evidence is accepted, refused, and rendered into claims (check_claims.py). Its object is the reporting system. A favorable software outcome establishes behavior of that control under its fixture; it does not establish the mechanics of any CAD or authorize an unscoped public safety statement.

4) build.dolfinx: Separate-Solver Capability (battery-build.dolfinx): The retained column references shared solver-comparison evidence (evidence/cad_battery/dolfinx_cross_check.json). It indicates the scope of that shared capability record, not a fresh dual-solver computation for every subject. Candidate-specific comparison requires the two actual execution identities, matched problem specification, and observable correspondence (G16-16).

5) geom.unit: Export Units (battery-geom.unit): The unit column (verify_genome_export.py) asks whether the exported representation's unit interpretation is established by the relevant evidence. All 23 historical cells are blocked. This does not prove that every export has incorrect units; it means the required binding was not established in those historical records.

6) geom.one_solid: Topology Binding (battery-geom.one_solid): This column (verify_genome_export.py) records whether single-solid topology was established for the subject. Its 23 blocked cells preserve a missing-evidence condition rather than 23 unfavorable topology measurements. A mesh closure check and a B-rep solid check do not establish the same fact.

7) geom.mass: Mass Observation (battery-geom.mass): The mass column (battery.rs) has 19 retained genome observations and four blocked reference entries. For those genome entries, the historical value is a plate/certification mass observation, not a demonstrated STEP-versus-raster consistency comparison.

8) geom.pin: Geometry Identity (battery-geom.pin): This column (cad_ingest.rs) concerns binding the observation to the intended CAD via geometry_sha256. All 23 historical entries are blocked because filenames alone cannot establish cryptographic identity.

9) plate.loads: Plate Response Finding (battery-plate.loads): The load column (fem.rs) reports the stored plate-response comparison: 19 unfavorable records and 4 blocked reference entries. Unfavorable findings are results of the model and comparison policy, not solver crashes; blocked entries are missing records, not measured failures.

10) plate.seq: Regional Response Comparison (battery-plate.seq): The sequence column (fem.rs) records 10 stored favorable and 9 unfavorable findings among genome entries (plus 4 blocked reference entries), inheriting the linear-elastic proxy limitations of G16-14.

11) plate.l7: Fatigue Screen (battery-plate.l7): The fatigue column (fem.rs) records 19 unfavorable model assessments and 4 blocked entries. It is a disclosed analytical screen, not a physical lifetime test.

12) plate.gci: Candidate Mesh Study (battery-plate.gci): This column (gci.py) reports candidate-specific convergence evidence: 4 favorable, 15 unfavorable, and 4 blocked records, distinct from the software arithmetic controls in build.gci_contracts.

13) side.a1: Shared Side-Report Reference (battery-side.a1): References a shared retained side report (g3_side_report.rs): 19 favorable references and 4 blocked entries, representing report availability under the historical producer's convention rather than a distinct executed experiment.

14) side.a2: Shared Side-Report Reference (battery-side.a2): Similarly references the common side report (19 favorable, 4 blocked); no distinct physical mechanism or separate execution is established solely by this row.

15) side.a3: Shared Side-Report Reference (battery-side.a3): Has the same evidential scope as side.a1 and side.a2 (19 favorable, 4 blocked). Treating A1, A2, and A3 as three independent successful experiments would manufacture corroboration from table structure.

16) side.a4: Support and Stability Report (battery-side.a4): Records the alternate-support/stability report (side_a4.py): 19 stored passes and 4 blocked entries. Report completeness, a favorable numerical finding, and physical support calibration remain distinct, and the eigsh $\mathbf{K}_s^{-1}\mathbf{G}$ symmetry/residual audit remains open.

17) solid.tip: Solid-Result Completeness (battery-solid.tip): Twenty stored passes and three blocked records (tip_solid.py). The 20 stored passes mean that local-solid result records were complete under the column's rules, not 20 favorable strength comparisons.

18) gate.rs_v2: Combined Grouped Finding (battery-gate.rs_v2): Records the combined historical RS-v2 finding (bench.rs): 19 failures and 4 blocked reference entries, with zero passes.

VII. Complete 44-Entry Catalog Crosswalk (G16, RSV2, and battery)

Every entry in docs/tests/catalog.json (16 G16 legacy predicates, 10 RSV2 requirement groups, and 18 battery columns = 44 total entries) is preserved in the machine-readable crosswalk (JR_EI_P3_T04_atlas_crosswalk_44.json) and summarized below with its exact docs/tests/README.md anchor, main-text Methodological Family (Section VIII, Families 1–11), and visual binding or explicit missing-visual disclosure:

- **Legacy G16 Diagnostics (16 entries):** g16-cad_watertight_and_single_solid (Family 1, export-identity.svg A–B); g16-geometric_and_holder_gates (Family 1, *subsection visual missing: geometry/reference visual not generated*); g16-karambit_ring_and_curved_claw (Family 1, *subsection visual missing: geometry/reference visual not generated*); g16-c6_aesthetic_floor (Family 2, reference.svg A–B); g16-cert_hiker_180n (Family 3, plate-fields.svg B); g16-cert_hunter_260n (Family 3, plate-fields.svg B); g16-strict_hunter_260n (Family 3, assessment.svg C); g16-cert_ring_pull_1500n (Family 4, plate-fields.svg C); g16-twist_body_1p82nm (Family 5, plate-fields.svg D); g16-twist_tip_only_1p82nm (Family 5, plate-fields.svg D); g16-twist_tip_submodel_strict_1p82nm (Family 5, assessment.svg C); g16-solid_tip_3d_p2_1p82nm (Family 6, verification.svg C–D); g16-side_a4_buckling_and_support (Family 7, assessment.svg A); g16-failure_sequence_1p2 (Family 8, verification.svg A); g16-plate_gci_valid_3level (Family 10, verification.svg A–B); g16-dolfinx_cross_check_2pct (Family 10, verification.svg C).
- **Grouped RS-v2 Predicates (10 entries):** rsv2-aesthetic_and_geometric (Family 2, reference.svg A–B); rsv2-floors_828_joint (Family 3/4, plate-fields.svg A–C); rsv2-twist_floor_body_and_tipsub_strict (Family 5, assessment.svg C); rsv2-solid_tip_sf (Family 6, verification.svg C–D); rsv2-strict_hunter_lateral (Family 3, assessment.svg C); rsv2-seq_hard (Family 8, verification.svg A); rsv2-human_max (Family 4/5, assessment.svg B); rsv2-fatigue (Family 9, assessment.svg B); rsv2-gate_s_sharp_law (Family 1, *subsection visual missing: exported-measurement visual not generated*); rsv2-gate_s_tip_point_brep (Family 1, *subsection visual missing: exported-measurement visual not generated*).
- **Historical Battery Evaluators (18 entries):** All 18 battery-* anchors map to battery-matrix.svg (exact evaluator-ID column across all 23 historical subjects), with secondary assessed-example visual links on battery-solid.tip (verification.svg C–D, Family 6), battery-plate.seq (verification.svg A, Family 8), battery-plate.gci (verification.svg A–B, Family 10), battery-side.a4 (assessment.svg A, Family 7), and battery-plate.l7 (assessment.svg B, Family 9).

VIII. Missing Subsection Visuals and Unavailable Spatial Fields

To prevent visual availability from being misread as full spatial-field coverage (‘40/44‘ catalog entries have subsection visual links; ‘6/44‘ link to the four retained $0.25\,\mathrm{mm}^2$ plate-field arrays in plate-fields.svg), we preserve two explicit non-availability ledgers from visuals.rs:

1) **Four Unpictured Geometry/Export Subsection Entries (subsection_visual_missing):**
   - g16-geometric_and_holder_gates and g16-karambit_ring_and_curved_claw: *“Subsection geometry/reference visual not generated; original flags are retained separately, without an admitted visual comparison.”*
   - rsv2-gate_s_sharp_law and rsv2-gate_s_tip_point_brep: *“Subsection exported-measurement visual not generated; nominal geometry and measured export observations must remain distinct.”*
2) **Four Unavailable Spatial Field Requests (status:archive_required):**
   - strict_plate_field: Only certification/strict scalar summaries ($168.9008\,\mathrm{N}$ vs. $168.1786\,\mathrm{N}$ lateral; 0.5239 vs. 0.5287 local torsion) are retained in the admitted publication input set; the $0.25\,\mathrm{mm}^2$ coarse plate field cannot substitute for a $0.0625\,\mathrm{mm}^2$ spatial array.
   - solved_solid_field: Only matched scalar safety-factor pairs and 1D longitudinal binned profiles are retained for the P_2 tetrahedral tip submodel; no full 3D volumetric stress array is archived in the publication inputs.
   - buckling_mode: Only the archived linear eigenvalue ($\lambda_1 = 7.0351$) and support schematic are retained; no spatial eigenmode displacement field is archived, and the eigsh $\mathbf{K}_s^{-1}\mathbf{G}$ operator-symmetry/residual verification question remains open.
   - fatigue_field: Only the saved modified-Goodman scalar ($n_f = 0.4718$) is retained; no per-element fatigue heatmap or physical cyclic life measurement exists in the admitted archive.

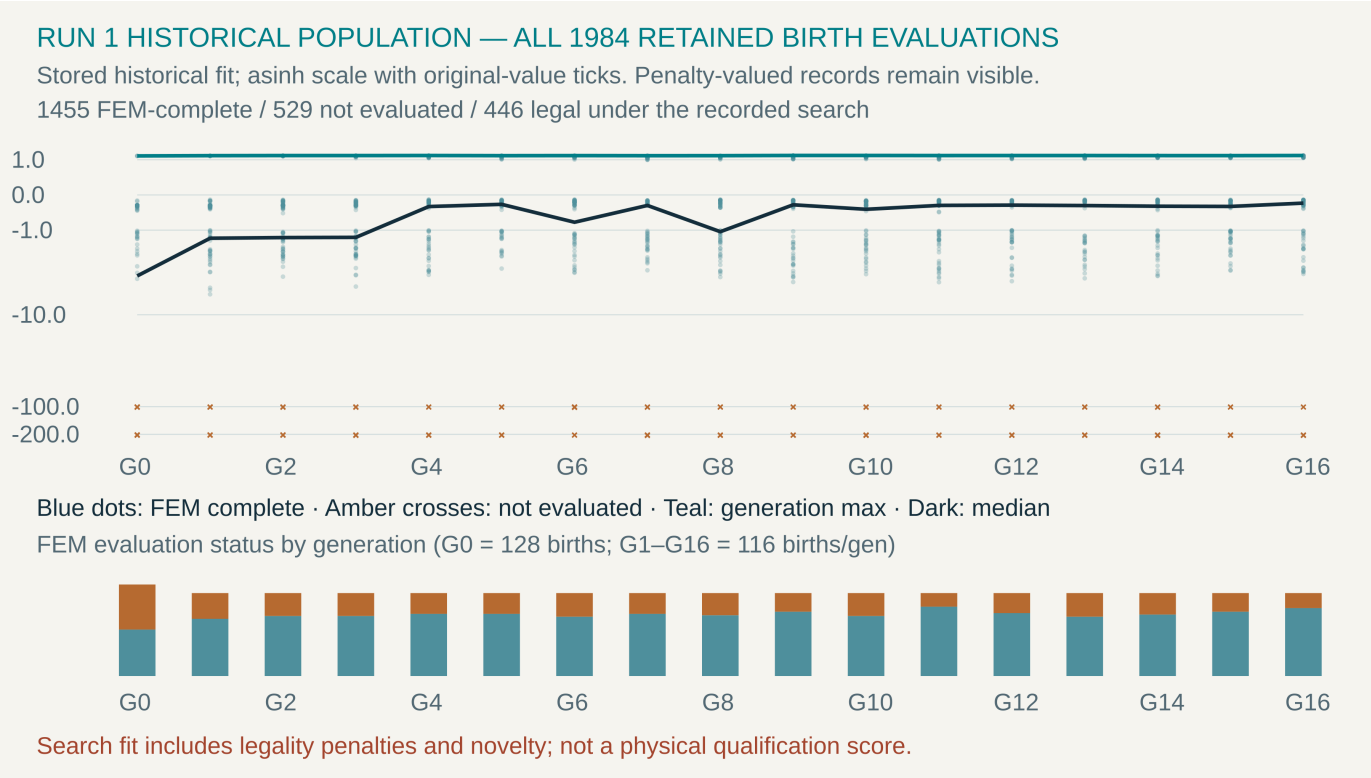


Fig. 2. Supplement Cohort 1 (Run 1) historical birth and island-membership records: all 1,984 retained births across generations 0–16 (1,455 FEM-complete blue points, 529 unevaluated amber crosses, 446 search-legal births) on an $\operatorname{asinh}(\text{fit})$ axis, plus 2,176 four-island slot memberships preserving 192 elite reappearances and 12 ring migrations at generations 4, 8, and 12.

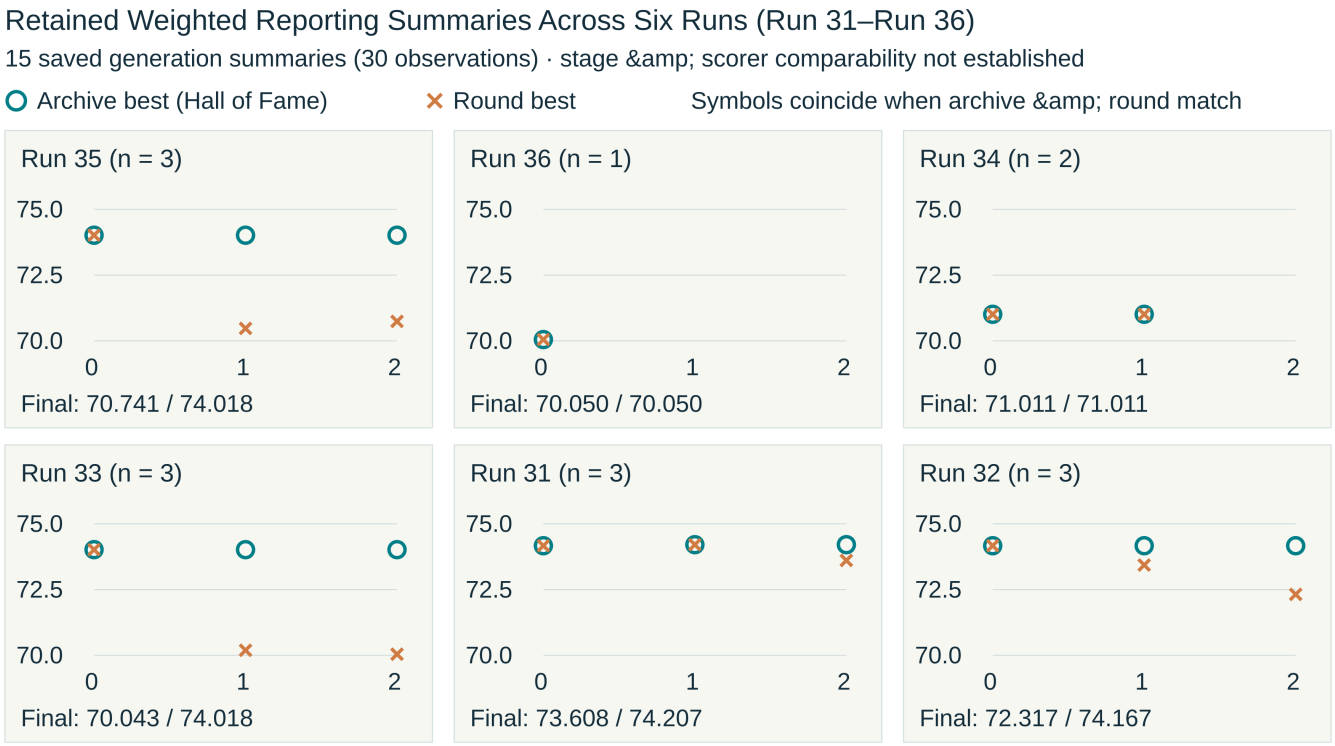


Fig. 3. Supplement Cohort 2 Retained Weighted Reporting Profile (Run 31–Run 36): all fifteen admitted generation summaries across six reporting series (70–75 vertical scale; hollow teal circles show stored archive values, orange crosses show round bests), peaking at $F = 74.207299$ for Individual 2002 in Run 31.

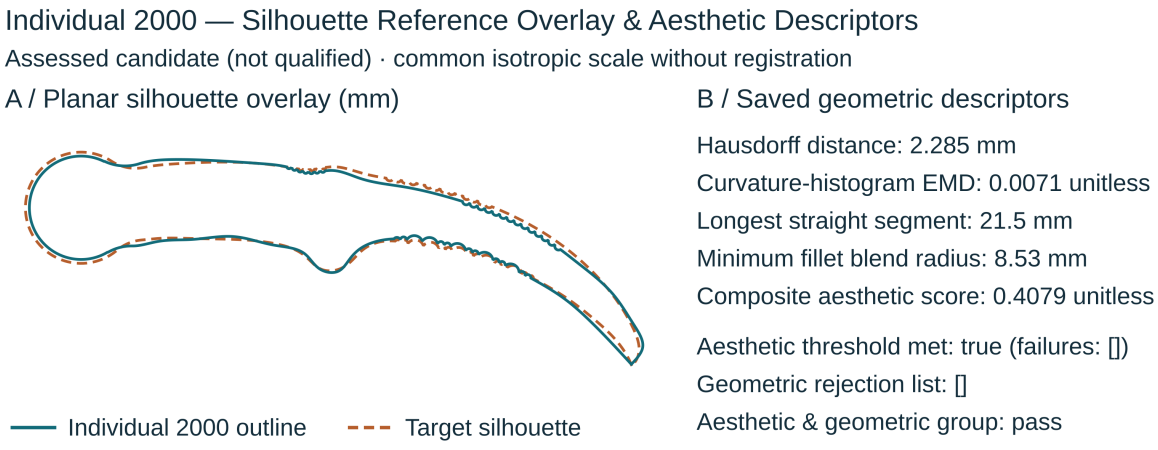


Fig. 4. Supplement Individual 2000 reference-contour comparison records (single assessed export, 1 assessed, 0 qualified): all 4,230 exterior outline points and 1,117 C.5/C.6 reference points displayed as closed polylines under a common isotropic scale, paired with saved descriptors (Hausdorff distance $2.285\,\mathrm{mm}$, curvature EMD 0.0071, similarity score 0.4079).

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